Sahaj Bhadja

(213) 284-4651 bhadja@usc.edu sahaj-bhadja sahajbhadja sahaj-bhadja

Summary

Infrastructure software engineer with 2+ years building **Java/Python/C++ services, distributed systems, and cloud-native applications**; experienced in Linux, storage, security, REST APIs, containers, CI/CD, automated testing, observability, performance, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Object-Oriented Programming, Software Engineering, Operating Systems, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **scalable backend services and REST APIs** across Python, FastAPI, PostgreSQL, SQL Server, Azure SQL, and cloud storage, supporting 500+ users while reducing latency from 1.2 seconds to 350 milliseconds.
- Engineered resilient highly available distributed processing with explicit states, failure isolation, validation, and recovery, improving throughput 3x while sustaining 99%+ reliability across 5,000+ monthly production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Developed **Python services and fault-tolerant automation** with multithreading, secure file ingestion, AWS RDS, and REST APIs, processing 1,000+ files per batch and 10,000+ daily requests.
- Integrated OpenAI, Claude, and Qwen behind adaptable application services, participating across requirements, design, implementation, testing, deployment, monitoring, production defect resolution, and support while reducing feedback turnaround 60%.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Built **secure cross-platform Java/Python microservices and integrations** across Linux, Windows, macOS, AWS Glue, and S3 storage, improving response times 30%, pipeline efficiency 68%, and storage costs 15%.
- Automated VM builds, installations, and 20+ unit and integration tests with Jenkins, PyTest, Mockito, Git, and CI/CD, generating coverage reports and failure logs for globally distributed teams.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **scalable secure distributed application infrastructure** across eight microservices, REST APIs, WebSockets, Azure SQL, and role-based access, synchronizing 2M+ records while maintaining modular architecture and data protection.
- Implemented Docker packaging, Jenkins delivery pipelines, zero-downtime releases, and automated testing while monitoring with Sentry, Azure Monitor, and Kibana, achieving 95%+ build validation and reducing deployment time 50%.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*

- Built **scalable cross-platform Flask services and data pipelines** for booking, transaction, and reporting workflows, integrating Firebase storage, Databricks, Spark, webhooks, and AWS-hosted components across 50,000+ records.
- Designed and documented 12+ system integrations with authentication, validation, and real-time synchronization, troubleshooting performance and customer issues while sustaining sub-200-millisecond API responses across mobile and web environments.

Skills

Languages: Java, Python, C++, JavaScript, TypeScript, SQL, Bash
Systems: Linux/Unix, OOP, data structures, distributed systems, multithreading, performance analysis, debugging
Cloud Infrastructure: Hybrid cloud, AWS, Azure, GCP, OpenShift, Kubernetes, Docker, storage, microservices
Reliability & Security: High availability, automated testing, DevSecOps, observability, monitoring, logging, access control
Delivery: REST APIs, SQL/NoSQL, Git, Jenkins, GitHub Actions, CI/CD, technical design, documentation, Agile

Projects

Chat2SQL Code | *Java, Linux, Docker, Data Systems*

- Enhanced a **reliable secure enterprise-class cross-platform application** supporting 10+ SQL and NoSQL systems with schema exploration, structured execution, storage workflows, Docker, Maven, Node.js, validation, and Linux-compatible debugging.

CPU Scheduling Simulator Live | *C++, Algorithms, Operating Systems*

- Implemented and validated **six C++ scheduling algorithms** with interactive controls, Gantt visualization, edge-case testing, and dynamic performance metrics across 1,000+ process combinations, reducing average turnaround time 25% and debugging time 20%.

Big-Brain Live | *AI, Cloud Storage, PostgreSQL*

- Engineered a **cloud data-ingestion and indexing system** processing 50,000+ records with secure storage, incremental updates, PostgreSQL metadata, validation, REST APIs, semantic retrieval, and performance tuning, reducing retrieval latency 40%.